

DECODING CUSTOMER VALUE

A SQL-Driven Retention Strategy

Full EDA + SQL Analysis Report
3,900 Customers | 18 Variables | 4 EDA Blocks | 5 SQL Analyses

METRIC	VALUE	SIGNAL
Total Customers	3,900	Full dataset, 37 rating nulls
Avg Value Score	38.5 / 100	Compressed — formula reweighted
Genuinely Loyal	40.3%	Buy at full price, no deals
Discount Hunters	11.6%	Low purchase history, high deal usage
Discount Spend Uplift	−\$0.85	Discounts reduce average spend
Subscriber Promo Rate	100%	Every subscriber uses a promo

Executive Summary

This report presents a complete analysis of a D2C fashion brand's customer behavioural dataset covering 3,900 customers across 18 variables. The analysis spans purchase history, discount and promo usage, subscription status, frequency of purchase, review ratings, geographic location, and a composite Customer Value Score. The EDA is complemented by five targeted SQL analyses that operationalise the findings into actionable customer profiles.

CORE FINDING

The brand is spending money on discounts that do not increase revenue. Discounts are being given to customers who would have purchased anyway — at the same price or higher. The subscription programme has been converted into a pure discount-delivery mechanism, eroding margin without creating loyalty. Meanwhile, 40.3% of customers already buy at full price and represent the brand's most valuable and underserved asset.

3,900	43%	40.3%	-\$0.85	100%
Total Customers	Promo Rate	Genuinely Loyal	Discount Uplift	Subscriber Promo Rate

Block 1 — Discount & Promo Dependency

The first block investigates how deeply discounts and promo codes are embedded in customer purchasing behaviour. The central question: are promotions building loyalty, or just attracting one-time bargain buyers?

1.1 Overall Promo & Discount Rates

Metric	Rate
Overall promo code usage	43.0%
Overall discount applied	43.0%
Promo rate — Subscribers	100.0%
Promo rate — Non-subscribers	21.9%

The promo and discount rates are identical at 43% — this is not a coincidence. Promo codes and discounts are applied together on almost every transaction. They are effectively the same lever.

INFERENCE 1.1

43% of all transactions involve a discount. This is very high for a D2C brand and suggests the brand has trained customers to wait for deals before purchasing. Customers no longer perceive the listed price as the real price.

1.2 Promo Usage Across Purchase Frequencies

Frequency	No Promo %	Promo Used %
Weekly	56.6%	43.4%
Bi-Weekly	58.7%	41.3%
Monthly	57.1%	42.9%
Quarterly	57.2%	42.8%
Annually	58.4%	41.6%

INFERENCE 1.2 (CRITICAL)

Promo usage is flat at ~43% across ALL frequency groups — weekly buyers use promos at almost the same rate as annual buyers. Frequent buyers have not been converted away from discount dependency. The brand is rewarding its most loyal customers with discounts they do not need.

1.3 Promo Usage — Subscribers vs Non-Subscribers

Subscription Status	No Promo %	Promo Used %
Subscribed (Yes)	0.0%	100.0%
Not Subscribed (No)	78.1%	21.9%

INFERENCE 1.3 (CRITICAL — STRUCTURAL PROBLEM)

Every single subscriber uses a promo code — 100%. The subscription programme is a discount access programme, not a loyalty programme. Non-subscribers use promos only 21.9% of the time, meaning organic buyers exist but they are not being converted to subscribers. A healthy subscription programme should create promo-independent loyalty. Instead, the brand has built a system where subscribing = guaranteed discount, funding the programme entirely through margin erosion.

1.4 Does a Discount Actually Increase Spend?

Discount Status	Avg Spend (USD)	Difference
No Discount Applied	\$60.13	—
Discount Applied	\$59.28	-\$0.85

INFERENCE 1.4 (THE MOST IMPORTANT NUMBER)

Customers who receive a discount spend \$0.85 LESS than those who do not. Discounts are not increasing basket size, order value, or revenue in any way. The brand is giving away margin for zero commercial return. This single number should inform an immediate review of the entire promotional architecture.

1.5 Discount Rate Across Categories

Category	Discount Applied %
Outerwear	44.4%
Accessories	43.8%
Footwear	43.2%
Clothing	42.1%

Discounts are uniformly distributed across all categories. No single category is driving the discount dependency. This rules out a category-specific problem and confirms it is a brand-wide pricing behaviour.

Block 2 — Loyalty & Retention

Block 2 examines how loyal customers actually are — measured by previous purchase counts, subscription status, review ratings, and the resulting segment classification.

2.1 Previous Purchase Distribution

Statistic	Value
Mean previous purchases	25.35
Median previous purchases	25.00
Std deviation	14.45
Minimum	1
25th percentile	13
75th percentile	38
Maximum	50

INFERENCE 2.1

The average customer has purchased 25 times before — a strong retention baseline. Customers are not churning after 1–2 purchases. The distribution is roughly uniform from 1 to 50, suggesting no natural drop-off point in purchase history.

2.2 Purchase History by Frequency

Frequency	Avg Previous Purchases
Weekly	25.77
Bi-Weekly	24.79
Monthly	25.28
Quarterly	26.85
Annually	24.56

INFERENCE 2.2

Purchase history is completely flat across frequency groups — weekly and annual buyers have almost identical purchase counts (~25). Frequency of purchasing does not predict total lifetime purchases. The brand does not have a 'high frequency = high value' dynamic, which is unusual and suggests purchase frequency is driven by habit rather than engagement.

2.3 Subscriber vs Non-Subscriber Comparison

Metric	Subscribers	Non-Subscribers	Difference
Count	1,053 (27%)	2,847 (73%)	—
Avg previous purchases	26.1	25.1	+1.0

Avg spend per transaction	\$59.49	\$59.87	-\$0.38
Avg review rating	3.75	3.75	0.00

INFERENCE 2.3

Subscribers and non-subscribers are virtually identical on every metric — same spend, same ratings, only 1 extra purchase on average. Subscription is not differentiating loyal customers. Combined with the 100% promo usage finding, this confirms the subscription programme has zero loyalty function — it is purely a discount delivery mechanism.

2.4 Review Ratings Across Frequencies

Frequency	Avg Review Rating
Weekly	3.76
Bi-Weekly	3.71
Monthly	3.77
Quarterly	3.73
Annually	3.76

INFERENCE 2.4

Review ratings are extremely flat — all frequency groups rate between 3.71 and 3.77. Nobody is particularly delighted or disappointed regardless of how often they buy. This signals a brand that is competent but not exceptional — customers are satisfied enough to keep buying but not inspired enough to advocate for it.

2.5 Customer Segment Breakdown

Segment	Count	% of Total	Definition
Loyal — No Deals	1,571	40.3%	15+ purchases, no promo/discount ever used
Loyal — Deal User	1,223	31.4%	15+ purchases, uses at least one promo/discount
Low Engagement	652	16.7%	Under 15 purchases, no deals
Discount Hunter	454	11.6%	Under 15 purchases, uses deals

INFERENCE 2.5

40.3% of customers are genuinely loyal buyers who never use deals — the brand's most valuable asset: 1,571 customers buying at full price repeatedly. The 11.6% Discount Hunters (454 customers) are the problem segment — low purchase history but high deal usage means the brand is acquiring and re-acquiring them at a loss.

Block 3 — Spend & High-Value Customer Profile

Block 3 investigates what a high-value customer looks like — which categories, age groups, geographies, and payment methods are associated with higher spend.

3.1 Spend by Category

Category	Avg Spend (USD)
Footwear	\$60.26
Clothing	\$60.03
Accessories	\$59.84
Outerwear	\$57.17

INFERENCE 3.1

Spend is almost completely flat across categories — highest (Footwear \$60.26) and lowest (Outerwear \$57.17) are only \$3.09 apart. Product category does not drive spend behaviour. Every category is being bought at the same average price point regardless of intrinsic value differences.

3.2 Spend by Age Group

Age Group	Avg Spend (USD)
18–25	\$60.65
46–55	\$60.58
26–35	\$59.76
36–45	\$59.31
55+	\$59.06

INFERENCE 3.2

Age has almost no impact on spending. The youngest group (18–25) spends the most at \$60.65 but the difference across all age bands is only \$1.59. The brand cannot use age as a targeting lever for higher-value purchases.

3.3 Spend by Payment Method

Payment Method	Avg Spend (USD)
Debit Card	\$60.92
Credit Card	\$60.07
Bank Transfer	\$59.71
Cash	\$59.70
PayPal	\$59.25

Venmo	\$58.95
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INFERENCE 3.3

Debit card users spend the most (\$60.92) and Venmo users the least (\$58.95) — a \$1.97 difference. Essentially flat. Payment method does not signal purchase intent or basket size for this brand.

3.4 Top 15 States by Avg Spend

State	Avg Spend (USD)
Alaska	\$67.60
Pennsylvania	\$66.57
Arizona	\$66.55
West Virginia	\$63.88
Nevada	\$63.38
Washington	\$63.33
North Dakota	\$62.89
Virginia	\$62.88
Utah	\$62.58
Michigan	\$62.10

INFERENCE 3.4

Geography shows the most meaningful spend variation — Alaska (\$67.60) vs the lowest states (~\$55–58) is a ~\$12 spread. Alaska, Pennsylvania, and Arizona are the highest-value states. Notably, the top 25% spenders (those spending \$84+) have a 42.3% promo usage rate — almost identical to the overall average of 43%. Even the highest-spending customers are discount-dependent.

3.5 The Pricing Ceiling Problem

Across every dimension analysed — category, age, payment method, geography, frequency — average spend per transaction hovers between \$57 and \$68. This is structural, driven by three overlapping forces:

Force 1 — Discount Anchoring

The discount programme has anchored customer price expectations at a discounted level. When 43% of all transactions involve a discount, customers learn the full price is not the real price. They wait. They use promo codes. The effective average transaction price gets pulled to a narrow band regardless of what they buy.

Force 2 — Flat Product Pricing

The product mix is relatively homogeneous in price point — clothing, accessories, footwear, and outerwear are all priced similarly. There is no premium tier or aspirational category pulling average order value up. A brand with a genuine premium segment would show category spend divergence of \$20–40+, not \$3.

Force 3 — Undifferentiated Targeting

No demographic segment shows meaningfully higher willingness to pay. The brand has not identified or cultivated a high-value customer archetype — everyone is treated the same and everyone spends roughly the same.

STRUCTURAL CONCLUSION

The combined result is a revenue ceiling. The brand can acquire more customers but each additional customer will spend ~\$59–60 per transaction. To grow revenue meaningfully, the brand must either increase transaction frequency (already attempted via subscriptions — not working) or break the pricing ceiling by reducing discount dependency and introducing premium tiers.

Block 4 — Customer Value Score (0–100)

The Customer Value Score is a composite metric that ranks all 3,900 customers on a single 0–100 scale by combining purchase volume, loyalty, frequency, satisfaction, and subscription status while penalising discount dependency.

4.1 Score Formula

Component	Variable	Weight	Direction	Reasoning
Spend	Purchase Amount (USD)	25	Positive	Higher spend = higher value
Purchase History	Previous Purchases	30	Positive	More orders = more loyal
Frequency	Frequency of Purchases	20	Positive	More frequent = more engaged
Satisfaction	Review Rating	15	Positive	Higher rating = more satisfied
Subscription	Subscription Status	10	Positive	Subscribed = more committed
Promo Penalty	Promo Code Used	–5	Negative	Promo user = discount dependent
Discount Penalty	Discount Applied	–5	Negative	Discounted = margin erosion risk

Continuous variables (Spend, Previous Purchases, Review Rating, Frequency) were normalised to a 0–1 scale using MinMaxScaler so no single variable dominates due to its raw numeric range. The final score is clipped to 0–100. Note: the formula shown reflects the recommended reweighting (reduced penalties from –7.5 to –5, purchase history increased from 20 to 30).

4.2 Score Distribution Results

Statistic	Value
Mean score	38.5 / 100
Median score	38.7 / 100
25th percentile	28.5
75th percentile	48.7
Maximum score	82.5
Minimum score	0.0
Std deviation	14.44

INFERENCE 4.1

The mean and median are almost identical at ~38.5, meaning the distribution is symmetrical. The majority of customers sit between 28.5 and 48.7 — a 20-point band. This compression makes it hard to distinguish top customers from average ones, which is the reason behind the formula reweighting.

4.3 Score by Subscription Status

Subscription Status	Median Value Score
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Not Subscribed	39.1
Subscribed	37.1

INFERENCE 4.2 (CONTRADICTION DETECTED)

Subscribers score LOWER than non-subscribers. This should not happen — subscription should indicate loyalty. The reason: since 100% of subscribers use promo codes, the promo penalty wipes out the subscription bonus. The recommended formula fix (reducing penalties from –7.5 to –5) partially corrects this while preserving the signal.

4.4 Score by Segment

Segment	Median Score	Interpretation
Loyal — No Deals	45.1	Correctly highest — full-price loyal buyers
Loyal — Deal User	36.0	Penalised for deal use despite high purchase history
Low Engagement	35.6	Low history, no deals — not rewarded enough
Discount Hunter	26.9	Correctly lowest — low history, deal dependent

INFERENCE 4.3

The score correctly separates Loyal No-Deal customers (45.1) from Discount Hunters (26.9) — an 18-point gap. The middle two segments remain close (36.0 vs 35.6), indicating further formula refinement is needed to separate loyal deal users from disengaged non-deal customers.

4.5 Recommended Formula Reweighting

Component	Current Weight	Recommended	Change
Purchase Amount	30	25	–5 (less weight on single transaction)
Previous Purchases	20	30	+10 (loyalty over time matters most)
Frequency	20	20	No change
Review Rating	15	15	No change
Subscription	10	10	No change
Promo Penalty	–7.5	–5	Reduce (subscribers over-penalised)
Discount Penalty	–7.5	–5	Reduce (same reason)

Block 5 — SQL Analysis

Five targeted SQL queries were written against the scored customer dataset to operationalise the EDA findings. Each query was designed to answer a specific strategic question.

Q1 — High vs Low Value Customer Profile

Value tiers were defined as: High Value (score ≥ 60), Mid Value (40–59), Low Value (<40). The query computed count, avg spend, avg purchases, avg score, promo rate, and subscription rate per tier.

Key findings from Q1:

High Value customers have the lowest promo rate (~30–35%) and the highest average previous purchases, confirming that true loyalty is inversely correlated with discount dependency. Mid Value customers represent the largest opportunity — they have reasonable purchase history but elevated promo usage, making them candidates for a promo-weaning programme. Low Value customers show the highest promo dependency, supporting the Discount Hunter characterisation.

SQL INSIGHT Q1

The single clearest differentiator between High and Low Value customers is not spend per transaction (compressed across tiers) but promo rate and purchase history. High Value customers have bought more times and use promos less — exactly what the segment model predicted.

Q2 — Promo Dependency by Segment

The query computed promo rate, discount rate, avg spend, avg purchase history, and avg value score for each of the four customer segments.

Segment	Promo Rate	Disc Rate	Avg Spend	Avg Purchases	Avg Score
Discount Hunter	100%	100%	~\$59	~8	~27
Loyal — Deal User	100%	100%	~\$60	~29	~36
Low Engagement	0%	0%	~\$60	~8	~36
Loyal — No Deals	0%	0%	~\$60	~30	~45

SQL INSIGHT Q2

The SQL confirms a binary promo structure: a customer either uses promos on every transaction or never uses them. There is no 'occasional promo user' middle ground. This binary pattern confirms the discount programme is not being used as a targeted retention tool — it is a structural feature of certain customer types.

Q3 — Geographic Opportunity Map

States were profiled on two dimensions — avg spend vs promo rate — and classified into four geo profiles. The threshold was avg spend $\geq \$62$ and promo rate $<40\%$ for 'Organic High Value'.

Geo Profile Definitions:

Profile	Criteria	Action
Organic High Value	Spend $\geq \$62$ AND promo rate $<40\%$	Protect and upsell
High Spend but Promo Driven	Spend $\geq \$62$ AND promo rate $\geq 40\%$	Wean off promos

Organic Low Spend	Spend <\$62 AND promo rate <40%	Improve AOV
Discount Driven	Spend <\$62 AND promo rate ≥40%	Re-evaluate investment

SQL INSIGHT Q3

Alaska, Pennsylvania, and Arizona emerge as the clearest Organic High Value geographies — high spend combined with relatively lower promo dependency. These states should receive first-priority treatment in any full-price campaign rollout. States in the 'High Spend but Promo Driven' quadrant (high spend, high promo rate) represent recoverable margin — customers who already spend at a premium level and may be convertible to full-price purchase behaviour.

Q4 — Category: Entry Point vs Retention

The query measured the share of new customers (<15 purchases) vs loyal customers (≥15 purchases) within each category, alongside promo rates.

Category	Avg Purchase History	New Customer %	Loyal Customer %	Promo Rate
Accessories	~25.4	~28%	~72%	43.8%
Clothing	~25.3	~28%	~72%	42.1%
Footwear	~25.2	~29%	~71%	43.2%
Outerwear	~25.0	~30%	~70%	44.4%

SQL INSIGHT Q4

Category loyalty profiles are nearly identical — all categories have ~70–72% loyal customer share and near-identical promo rates. Outerwear has marginally more new customers and the highest promo rate (44.4%), suggesting it may serve as an entry-category. However the differences are too small to drive category-specific strategy. The uniform distribution again confirms this is a brand-wide issue rather than a category problem.

Q5 — Ideal Customer Profile (ICP)

The query filtered only Loyal — No Deals customers and ranked every combination of Gender, Age Group, Category, Frequency, Payment Method, and Season by average Value Score. The top 15 profiles were extracted.

Top ICP characteristics (top-ranked profiles):

- Male customers in the 46–55 and 55+ age groups consistently appear at the top of the ranking
- Quarterly purchase frequency dominates the top profiles — not weekly
- Footwear and Clothing are the most common categories in high-scoring profiles
- Debit Card and Bank Transfer are the most frequent payment methods
- Fall and Winter seasons index highest in the top profiles
- Zero promo usage across all ICP profiles by construction — these customers never needed a deal

SQL INSIGHT Q5 — THE IDEAL CUSTOMER

The ICP is a 46–55 year old male customer buying Footwear or Clothing quarterly, paying by Debit Card or Bank Transfer, with purchases concentrated in Fall/Winter. This profile has the highest value score, the longest purchase history, and zero reliance on discounts or promos. This is who the brand should be targeting in acquisition campaigns, and whose behaviour should be used to redesign the subscription and loyalty programme.

Strategic Recommendations

R1 — Dismantle the discount-as-subscription model

Decouple promo codes from subscription status immediately. The programme currently guarantees a discount to every subscriber, which means the brand pays for every subscription renewal with margin. Redesign the subscription programme around early access, loyalty points, or exclusive products rather than automatic price reductions.

R2 — Protect the 40.3% Loyal — No Deals segment

These 1,571 customers buy at full price repeatedly and should never receive unsolicited discount outreach. Create a protected tier (e.g. 'Brand Loyalist') that routes them to premium experience tracks: early product access, priority service, community features. Any discount sent to this group is pure margin loss.

R3 — Run a promo-weaning programme on Loyal — Deal Users

The 1,223 Loyal Deal Users have strong purchase history but 100% promo dependency. These customers have already demonstrated willingness to buy repeatedly — they do not need the discount, they have just been trained to expect it. A staged reduction in promo offer frequency (50% → 25% → 0% over 6 months) with substitution of non-monetary value (bonus product, personalised content) should be tested on a holdout group.

R4 — Immediately cut acquisition spend on Discount Hunters

The 454 Discount Hunters (11.6%) cost the brand money: low purchase history means the acquisition cost is not recovered, and 100% promo usage means every transaction erodes margin. These customers should not be targeted in acquisition campaigns, and retargeting spend on lapsed Discount Hunters should be paused.

R5 — Introduce a genuine premium tier to break the pricing ceiling

With all current spend compressed between \$57–68, the only path to higher average order value is a new product tier at a meaningfully higher price point (\$100–150 range). The ICP analysis (Q5) identifies the profile most likely to convert to a premium tier: male, 46–55, Footwear/Clothing, quarterly buyer. This cohort should receive first access to any premium range testing.

R6 — Prioritise Alaska, Pennsylvania, and Arizona for organic growth campaigns

These three states demonstrate the highest spend and the lowest promo dependency — the brand already has organic buyers here at scale. Full-price campaign investment in these geographies will yield higher ROI than the same spend in discount-driven markets.

Appendix — Methodology & Data Notes

Dataset Overview

Attribute	Detail
Total records	3,900 customers
Variables	18 columns including Customer ID, Age, Gender, Location, Category, Item Purchased, Size, Color, Purchase Amount
Missing values	37 null values in Review Rating — filled with median (3.75) for scoring; excluded from rating-specific analyses
Cleaning applied	'Fortnightly' merged into 'Bi-Weekly'; 'Every 3 Months' merged into 'Quarterly' to eliminate duplicate frequency labels
Tools used	Python (pandas, scikit-learn, matplotlib, seaborn), SQLite3 (in-memory), Power BI for dashboard visualisation

Variable Score Normalisation

MinMaxScaler was applied to Purchase Amount (USD), Previous Purchases, Frequency (mapped to ordinal: Weekly=5, Bi-Weekly=4, Monthly=3, Quarterly=2, Annually=1), and Review Rating before score computation. This ensures the score reflects relative position within the dataset rather than absolute values, making it robust to future dataset updates.

Segment Definition Logic

Segment	Condition
Loyal — No Deals	Previous Purchases ≥ 15 AND Promo Code Used = No AND Discount Applied = No
Loyal — Deal User	Previous Purchases ≥ 15 AND (Promo Code Used = Yes OR Discount Applied = Yes)
Low Engagement	Previous Purchases < 15 AND Promo Code Used = No AND Discount Applied = No
Discount Hunter	Previous Purchases < 15 AND (Promo Code Used = Yes OR Discount Applied = Yes)

The 15-purchase threshold for loyalty designation was chosen to represent roughly the 60th percentile of purchase history. Customers above this threshold have demonstrated repeated engagement beyond the early trial phase.